



Withholding matters: The impact of Act 32 on compliance with the earned income tax[☆]

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ABSTRACT

This paper examines Act 32 of the Pennsylvania state legislature which mandated the introduction of withholding for the local earned income tax (EIT) for all employees and the consolidation of a fragmented collection system to one collector per county effective January 1, 2012. The estimates I obtain suggest that the act resulted in increased compliance with the EIT of about 13 percent. A falsification exercise examining compliance with the property tax for the identical municipalities confirms that Act 32 did not impact the property tax. Both findings are robust to the inclusion of municipality fixed effects or municipality-specific time trends.

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1. Introduction

A growing literature in public economics emphasizes the importance of information reporting and withholding for tax compliance. Withholding by employers plays a particularly critical role in ensuring higher compliance. Withholding combines third-party reporting by which the revenue agency can cross-check an employee's self-reports with information reported by employers and the act of withholding itself, i.e. collecting the tax at the source rather than from the recipient of the income.

In an attempt to spur higher tax compliance, governments routinely enact policies that bring more activities under the ambit of information reporting and withholding. This paper studies one such reform: Act 32 of 2008 that completely restructured the collection of local earned income taxes in the state of Pennsylvania. Pennsylvania is relatively unique in that most of its municipalities impose a local earned income tax (EIT) and that tax accounts for about half of all taxes collected. Signed into law on July 2, 2008, Act 32 provided that by January 1, 2012, employers would be required to withhold the EIT on behalf of all their employees. The act furthermore reduced the number of tax collectors from approximately 560 to 69, i.e. one tax collector for each of the 66

counties, with the only exception being Allegheny County that was split into four districts.¹

The question that I answer in this paper is: What were the causal effects of this policy reform – the state-mandated introduction of withholding for all employees and the consolidation of tax collectors – on local EIT revenues? Using a panel dataset of 2,318 municipalities and an 8-year window around the implementation of Act 32, I find that withholding led to an immediate and permanent increase in compliance by about 13 percent. These results are robust to the inclusion of municipality fixed effects or municipality-specific linear time trends. To address the concern that an improvement in local economic conditions may have led to this increased compliance, I conduct a falsification exercise examining compliance with the property tax for the identical set of 2,318 municipalities and find that Act 32 had *no* impact on the collection of this tax.

This paper contributes to a literature that documents the importance of withholding for tax compliance (e.g., Slemrod and Velayudhan, 2018; Waseem, 2022). This literature finds increases in tax collections through a default payment channel and change in enforcement perceptions (Brockmeyer and Hernandez, 2019), by reducing frictions in the remittance of taxes (Boning, 2018), and through reduced non-filing (Bagchi and Dušek, 2021).

[☆] I greatly appreciate the comments and suggestions from Nathan Anderson, Traviss Cassidy, Libor Dušek, Jeffrey Hoopes, Christopher Kilby, Jagadeesh Sivadasan, and Joel Slemrod. All errors remain my own. All links were current as of July 27, 2022.

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¹ Although technically a county, Philadelphia does not have an independent county government. Philadelphia has its own provisions governing the EIT and was outside the purview of Act 32.

One potential concern about drawing inferences from some of these reform episodes is that they were introduced by governments that were the direct beneficiary of higher collections and it seems plausible that governments adopted these reforms when they were in dire need of higher revenues. Thus beyond introducing these reforms, governments may have increased the intensity of enforcement or made changes to the tax base and data on such changes are typically unavailable to researchers. In contrast, the reform I study in this paper – Act 32 of 2008 – was designed to improve compliance with the *local* EIT – a source of revenue relied upon by *local* governments, as opposed to the Commonwealth of Pennsylvania itself. Moreover, the date when these reforms would become effective (and the base of the tax itself) was set into law by the state legislature, with local governments having no discretion on those dimensions. Accordingly, the endogeneity concerns that might plague prior estimates of the introduction of withholding have less bite.

The rest of the paper is organized as follows. Section 2 provides institutional background on EIT collection practices. Section 3 describes the data and our empirical specification. I present the baseline results in Section 4.1, with possible mechanisms in Section 4.2. Section 5 concludes.

2. Institutional background

The EIT is a commonly used source of revenue for Pennsylvania's local governments accounting for about half of all municipal taxes and rivaling the property tax as the most important source of revenue. Income taxes have long been a staple of Pennsylvania with local governments authorized since 1965 to levy an EIT on wages, salaries, commissions, net profits or other compensation of residents. With limited exceptions, such as those for fiscal distress, the EIT rate is capped at 1 percent, with proceeds generally split equally between the municipality and the school district.

Despite the state's extensive experience with the EIT, the collection of this tax was not streamlined. Prior to Act 32, employers had been required to withhold the EIT only when employees lived and worked in the same municipality. Second, individual local governments were authorized to select a collector, resulting in a fragmented system with about 560 tax collectors statewide. Problems stemming from the lack of withholding and fragmentation included employees not filing local income taxes with the correct collector, some employees not filing at all, and collectors returning taxes paid by the employer in error, rather than forwarding those receipts to the correct tax officer leading to an estimated annual revenue loss of between \$100 to \$237 million (p. 4, [Legislative Budget and Finance Committee, 2016](#)).

In response Pennsylvania passed Act 32 in 2008 enacting numerous changes. Most importantly, following Act 32, businesses were required to withhold taxes for *all* employees and to ensure that those revenues reached the appropriate tax collectors. Given that fewer than 12 percent of all workers lived and worked in the same municipality on average (see Table A.1), Act 32 had the practical effect of introducing withholding for the EIT when none existed before. Second, Act 32 required the consolidation of tax collectors to one for each county (except Allegheny, which had four districts). The degree of consolidation achieved in practice was even more significant as two firms – Berkheimer Tax Administrator and Keystone Collections Group – emerged as collectors of choice and were chosen by 48 of the 69 taxing districts (p. 4, [Legislative Budget and Finance Committee, 2016](#)).

3. Empirical analysis

3.1. Construction of variables and data sources

Data regarding Pennsylvania's municipal finances – tax revenues by source and tax rates – are available annually from the

Pennsylvania Department of Community and Economic Development (DCED). To construct a measure of compliance, I need data on aggregate earnings, i.e. income from wages and salaries and self-employment income, received by households in a given municipality, which are drawn from the American Community Survey. I define the dependent variable as:² (see Eq. (3.1) in Box 1).

Summary statistics and data sources are outlined in Table A.1.

3.2. Empirical specification

I use an event study approach with an 8-year window around 2012, the first year in which Act 32 takes effect, to examine its impact on compliance with the EIT. I start off with a parsimonious specification that includes county fixed effects and a linear time trend and then add socioeconomic controls and the EIT rate. More stringent specifications include municipality fixed effects and, eventually, municipality-specific linear time trends. The most complete specification is:

$$Y_{mt} = \alpha + \beta_1 * Post_t + \beta_2 * T + \beta_3 * X_{mt} + \lambda_m + \lambda_m * T + \varepsilon_{mt} \quad (3.2)$$

where $Post_t$ is a dummy variable set to 1 for the years 2012 through 2015 and zero for the years 2008 through 2011; X_{mt} are time-variant controls at the municipal level, such as the log of median household income, the municipal EIT resident rate, and the average vote share of Democratic candidates in national and state elections. λ_m are municipality fixed effects, $\lambda_m * T$ are municipality-specific linear time trends, and ε_{mt} is the error term. I cluster standard errors at the county level given that this choice results in the most conservative standard errors.

4. Results

4.1. Baseline specification

Before presenting the regression results, a cross-tabulation and visual representation of the underlying data is helpful. Between 2011 and 2012, the median EIT compliance rate increases by about 7 percent. To the extent that annual data are noisy, I can construct an average for 4 years pre- and post-Act 32 and that suggests an increase in the compliance rate by about 9 percent. These increases are borne out in a plot of the compliance rate in Fig. 1. Finally, these patterns also emerge when I examine the evolution of aggregate EIT revenues collected by Pennsylvania municipalities vis-à-vis those in Ohio, alongside state personal income tax revenues for Pennsylvania in Fig. A.2. Although all of these snapshots of the data are unsophisticated and omit controls, subsequent regressions confirm these findings.

I present my baseline estimates in Panel A of Table 1. Column (1) is parsimonious and only includes a dummy variable for the post-Act 32 period, a linear time trend to account for any secular changes over time, and county fixed effects. Column (2) introduces municipality fixed effects and column (3) adds municipality-specific linear time trends. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but they now include numerous time-variant controls at the municipal level.

As the results in Panel A indicate, the implementation of Act 32 is associated with a statistically and economically significant increase in the EIT compliance rate of around 13 percent. Furthermore, these results are not sensitive to the time window employed. Table 2 confirms that the increased compliance manifests itself when constructing windows of varying length around the year of adoption, with suggestive evidence that the effects of Act 32 may have grown over time.

² The EIT does not have any personal exemption or standard deduction. It is a flat tax applied on all wages, salaries, commissions, net profits or other compensation. See Fig. A.1 for an example of an EIT form.

$$\text{Compliance with the EIT} = \frac{\text{Earned income taxes collected}}{(\text{EIT rate}/100 * \text{Aggregate Wages and Salaries} + \text{Self} - \text{employment income})} * 100 \quad (3.1)$$

Box 1.

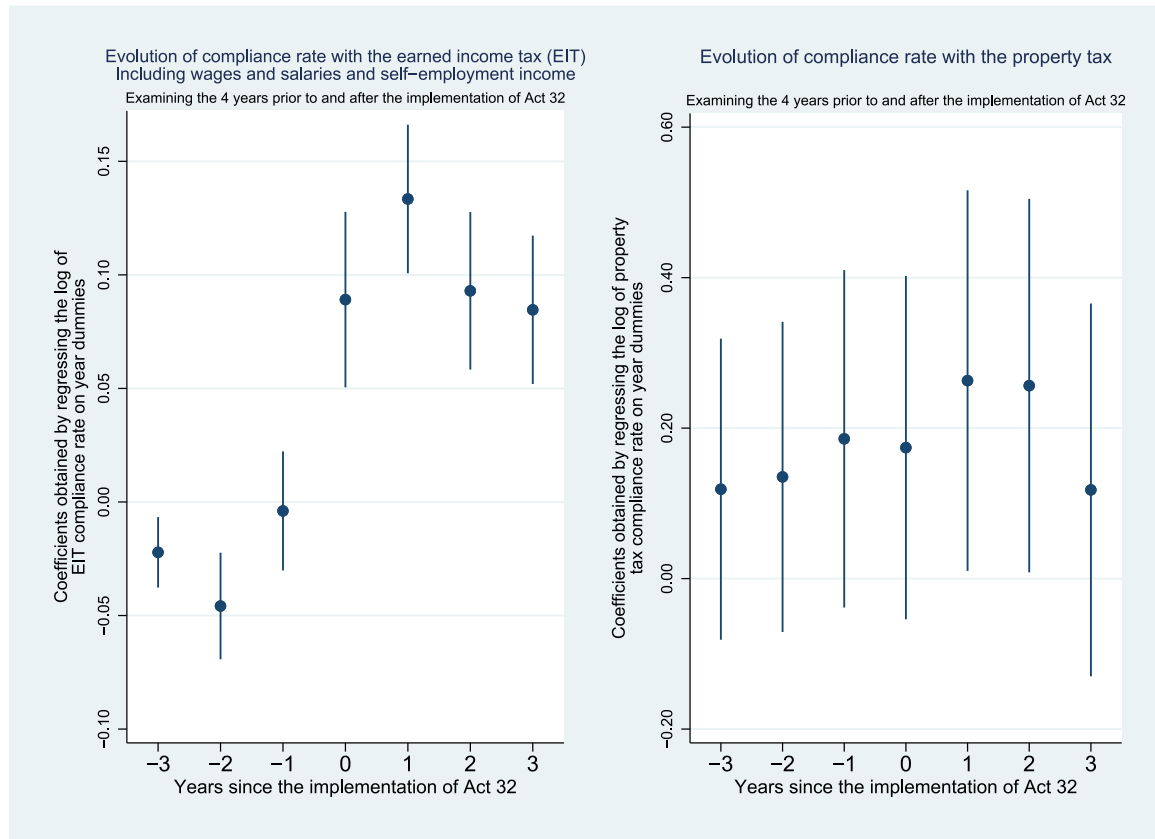


Fig. 1. Plot of compliance rate with the Earned Income Tax (EIT) and the property tax for an 8-year window around the year of adoption of Act 32. Notes: The left panel plots the coefficients, along with the 95 percent confidence intervals, from a regression of the log of compliance with the EIT on a set of year dummies, with standard errors clustered at the county level. The year 2008 is the omitted category and all of the coefficients are changes in compliance relative to the base year. The right panel replicates the exercise in the left panel but examines the log of compliance with the property tax.

Table 1

Baseline specification: Effects of the introduction of Act 32 on compliance using an 8-year window around the year of adoption.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dependent variable: Log of compliance rate with the earned income tax (EIT)						
Dummy variable for Post-Act 32 period	0.135*** (0.0150)	0.137*** (0.0150)	0.137*** (0.0161)	0.145*** (0.0190)	0.131*** (0.0181)	0.110*** (0.0201)
Number of observations	17863	17863	17863	17863	17863	17863
R ²	0.072	0.53	0.72	0.11	0.57	0.74
Panel B: Dependent variable: Log of compliance rate with the property tax						
Dummy variable for Post-Act 32 period	0.0127 (0.0340)	0.0176 (0.0344)	0.0175 (0.0371)	−0.0244 (0.0291)	−0.00527 (0.0161)	0.00712 (0.00783)
Number of observations	17863	17863	17863	17863	17863	17863
R ²	0.49	0.80	0.90	0.52	0.84	0.92
Socioeconomic controls included	No	No	No	Yes	Yes	Yes
Fixed effects introduced	County	Municipality	Municipality	County	Municipality	Municipality
Linear time trends introduced	Yes	Yes	Municipality-specific	Yes	Yes	Municipality-specific

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The socioeconomic controls included in cols. (4)–(6) in Panel A are the log of median household income, percentage of aggregate income from wages and salaries, the unemployment rate, percentage of housing units that are vacant, percentage of the population that is less than 18 years and older than 65 years, percentage of workers that live and work in the same municipality, the municipal EIT resident rate, and the average vote share of Democratic candidates in national and state elections. The identical set of controls are included in Panel B except that instead of controlling for the municipal EIT rate, I control for the municipal millage rate. The regressions are estimated on the set of municipalities that have both an earned income tax and a property tax in place and exclude Philadelphia which was outside the scope of Act 32.

Table 2
Robustness of results to varying the time window around the implementation of Act 32.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Examining a 6-year window around the year of adoption						
Dummy variable for Post-Act 32 period	0.113*** (0.0168)	0.116*** (0.0167)	0.112*** (0.0181)	0.117*** (0.0187)	0.116*** (0.0184)	0.0990*** (0.0210)
Number of observations	13436	13436	13436	13436	13436	13436
R ²	0.085	0.60	0.78	0.12	0.64	0.80
Panel B: Examining an 8-year window around the year of adoption (Baseline specification)						
Dummy variable for Post-Act 32 period	0.135*** (0.0150)	0.137*** (0.0150)	0.137*** (0.0161)	0.145*** (0.0190)	0.131*** (0.0181)	0.110*** (0.0201)
Number of observations	17863	17863	17863	17863	17863	17863
R ²	0.072	0.53	0.72	0.11	0.57	0.74
Panel C: Examining a 10-year window around the year of adoption						
Dummy variable for Post-Act 32 period	0.138*** (0.0137)	0.140*** (0.0137)	0.139*** (0.0144)	0.152*** (0.0196)	0.137*** (0.0177)	0.114*** (0.0191)
Number of observations	22335	22335	22335	22335	22335	22335
R ²	0.066	0.48	0.64	0.10	0.53	0.67
Panel D: Examining a 12-year window around the year of adoption						
Dummy variable for Post-Act 32 period	0.135*** (0.0136)	0.136*** (0.0136)	0.135*** (0.0140)	0.151*** (0.0190)	0.140*** (0.0163)	0.126*** (0.0181)
Number of observations	26772	26772	26772	26772	26772	26772
R ²	0.063	0.44	0.60	0.098	0.49	0.63
Socioeconomic controls included	No	No	No	Yes	Yes	Yes
Fixed effects introduced	County	Municipality	Municipality	County	Municipality	Municipality
Linear time trends introduced	Yes	Yes	Municipality-specific	Yes	Yes	Municipality-specific

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in all panels is the log of compliance with the earned income tax (EIT). The socioeconomic controls included in cols. (4)–(6) in all panels are the log of median household income, percentage of aggregate income from wages and salaries, the unemployment rate, percentage of housing units that are vacant, percentage of the population that is less than 18 years and older than 65 years, percentage of workers that live and work in the same municipality, the municipal EIT resident rate, and the average vote share of Democratic candidates in national and state elections. The regressions are estimated on the set of municipalities that have both an earned income tax and a property tax in place and exclude Philadelphia which was outside the scope of Act 32.

One possible concern with these results is that the pre-Act 32 period includes years impacted by the Great Recession and accordingly, the increased compliance may have resulted from an improvement in economic conditions rather than from the act itself. That concern is however misplaced because an improvement in economic conditions would go hand-in-hand with an increase in aggregate household earnings and the EIT compliance rate is already being defined in relation to those aggregates. Additionally, in cols. (4)–(6) I control for proxies of local fiscal stress by including the unemployment rate and the percentage of housing units that are vacant but doing so does not appreciably affect the estimates. Likewise, allowing the EIT compliance rate in each municipality to flexibly evolve over time does not impact our estimates either. Nonetheless, I conduct a falsification exercise in Panel B of Table 1 where I examine compliance with the *property tax* for the identical set of municipalities as considered in Panel A.³ In sharp contrast to the earlier results, I find that the implementation of Act 32 does *not* result in higher compliance with that tax, with two of the six point estimates being negative.

A final possible concern with these results demonstrating higher compliance with the EIT is that the improvement in compliance could be driven by variation in the measure of aggregate earnings used in the denominator of (3.1) rather than by increases in the earned income taxes collected. This concern can be addressed head-on by examining the effects of Act 32 on earned income tax revenues collected per capita. Those results presented in Fig. A.3 confirm the increased collection of the EIT following the adoption of Act 32, with the estimates in Table A.2

pointing to an increase of about 13 percent. Additional robustness checks using county-level data on taxable compensation and net profits from the Pennsylvania Department of Revenue confirm these results and they are available on request.

4.2. Mechanisms underlying increased compliance

One of the primary issues that Act 32 sought to address was the fragmented collection regime in place prior to 2012 by mandating a single collector for each county. In a check designed to shed light on the underlying mechanisms, I interact the reduction in number of collectors experienced by a county between 2010 and 2012 with the post-Act 32 dummy and re-estimate our specifications. Those results, presented in Table A.3, show that both aspects of Act 32 contributed to the increased compliance. Thus, while the introduction of withholding for all individuals not residing and working in the same municipality resulted in an increase in compliance of about 9 percent (using an 8-year window) or nearly three-fourths of the overall impact of 13 percent, counties with more fragmented systems prior to Act 32 experienced additional improvements following consolidation.

5. Conclusions

This study extends the literature that has highlighted the importance of third-party information reporting and withholding for tax compliance. Unlike prior work that has examined the endogenous adoption of reforms by jurisdictions that were the direct beneficiaries of additional revenues, this paper represents the first examination of a reform where such concerns around endogeneity are significantly attenuated. The date as of which these reforms came into effect was set into law by the Pennsylvania state legislature, with local governments having no discretion

³ Compliance with the property tax is defined as Property taxes collected/(Millage rate/1000 * Aggregate value of the housing stock in the municipality) * 100.

on whether to adopt these reforms or not. This paper suggests that the changes to the collection regime mandated by Act 32 nonetheless were material and led to a statistically and economically significant increase in compliance with the EIT of about 13 percent.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econlet.2022.110796>.

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Supplementary Material to: Public-Sector Pension Plans and the Discount
Rate Assumption: The Role of Political Incentives
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Table A.1: Summary Statistics

Variable	Units	Mean	Median	Standard deviation	Minimum	Maximum
Pension plan characteristics						
For municipal plans:						
Interest rate in 2003 (Unwinsorized)	In percent terms	7.04	7.00	0.682	5.00	9.00
Interest rate in 2003 (Winsorized as noted below)	In percent terms	7.04	7.00	0.660	5.50	8.00
Interest rate in 2013 (Unwinsorized)	In percent terms	6.66	7.00	1.019	3.00	8.50
Interest rate in 2013 (Winsorized as noted below)	In percent terms	6.69	7.00	0.958	5.50	8.00
Actuarial funded ratio	In percent terms	103.72	93.49	48.42	45.07	304.57
Number of active members		33.8	7	501.99	0	20540
Number of beneficiaries		33.2	3	593.10	0	23117
For plans run by municipal authorities:						
Interest rate in 2003 (Unwinsorized)	In percent terms	6.79	6.50	0.557	5.00	8.50
Interest rate in 2003 (Winsorized as noted below)	In percent terms	6.79	6.50	0.535	5.50	8.00
Interest rate in 2013 (Unwinsorized)	In percent terms	6.15	5.50	0.945	5.00	8.50
Interest rate in 2013 (Winsorized as noted below)	In percent terms	6.15	5.50	0.928	5.50	8.00
Actuarial funded ratio	In percent terms	104.73	98.43	36.60	45.07	304.57
Number of active members		21.2	9	37.95	0	361
Number of beneficiaries		9.16	2	24.38	0	212
Controls at the municipal level						
Unemployment rate	In percent terms	6.89	6.28	2.94	2.08	14.65
Debt per capita	In dollars	474.6	196.6	616.6	0	2135
Ratio of taxes to per capita income		1.20	1.08	0.55	0.36	2.60
Per capita income	In dollars	25510	23798	7816	14415	46643

Summary statistics for actuarial funded ratio, interest rate, Social security coverage, and number of active members and beneficiaries are based on biennial data from 2003–2013 from the PERC. Coverage under collective bargaining is based on the 1982 Employment Summary Statistics of Census of Governments. Unemployment rate and per capita income are from the 2007–11 and 2011–15 ACS. Debt per capita and ratio of taxes to per capita income are constructed based on annual data from the Pennsylvania Department of Community and Economic Development (DCED). Unless otherwise mentioned, all variables have been winsorized at the 2.5% and 97.5% levels.

Table A.2: Baseline specification: DiD effects of stronger incentives for elected officials on the discount rate chosen by pension plans

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Baseline specification – Winsorizing the dependent variable and all control variables at the 2.5% level						
Dummy variable for municipality	0.291***	0.281***	0.262***	0.299***	0.292***	0.274***
X Dummy variable for “Post-period”	(0.0391)	(0.0387)	(0.0382)	(0.0426)	(0.0419)	(0.0408)
Panel B: Without winsorizing the dependent variable and any of the control variables						
Dummy variable for municipality	0.291***	0.281***	0.262***	0.299***	0.292***	0.274***
X Dummy variable for “Post-period”	(0.0383)	(0.0399)	(0.0395)	(0.0384)	(0.0393)	(0.0389)
Panel C: Winsorizing the dependent variable and all of the control variables at the 1% level						
Dummy variable for municipality	0.268***	0.257***	0.244***	0.281***	0.273***	0.261***
X Dummy variable for “Post-period”	(0.0397)	(0.0397)	(0.0396)	(0.0431)	(0.0430)	(0.0427)
Panel D: Winsorizing the dependent variable and all of the control variables at the 5% level						
Dummy variable for municipality	0.291***	0.289***	0.261***	0.290***	0.288***	0.265***
X Dummy variable for “Post-period”	(0.0391)	(0.0396)	(0.0391)	(0.0434)	(0.0431)	(0.0415)

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: All results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups, e.g. non-uniformed personnel or policemen and firefighters. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income.

Table A.3: Baseline specification, augmented with a control for the change in level of assets between 2007 and 2009 as a proxy for the asset allocation

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Baseline specification – Winsorizing the dependent variable and all control variables at the 2.5% level						
Dummy variable for municipality	0.291***	0.281***	0.263***	0.293***	0.285***	0.268***
X Dummy variable for “Post-period”	(0.0392)	(0.0388)	(0.0381)	(0.0430)	(0.0423)	(0.0410)
Panel B: Without winsorizing the dependent variable and any of the control variables						
Dummy variable for municipality	0.263***	0.263***	0.262***	0.272***	0.272***	0.272***
X Dummy variable for “Post-period”	(0.0400)	(0.0405)	(0.0406)	(0.0443)	(0.0446)	(0.0446)
Panel C: Winsorizing the dependent variable and all of the control variables at the 1% level						
Dummy variable for municipality	0.268***	0.257***	0.245***	0.274***	0.266***	0.255***
X Dummy variable for “Post-period”	(0.0398)	(0.0398)	(0.0396)	(0.0434)	(0.0434)	(0.0430)
Panel D: Winsorizing the dependent variable and all of the control variables at the 5% level						
Dummy variable for municipality	0.291***	0.289***	0.262***	0.285***	0.283***	0.261***
X Dummy variable for “Post-period”	(0.0392)	(0.0397)	(0.0387)	(0.0439)	(0.0435)	(0.0417)

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: All results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups, e.g. non-uniformed personnel or policemen and firefighters, along with the change in level of assets between 2007 and 2009 as a proxy for the asset allocation. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income.

Table A.4: Robustness checks confirming the differential response of discount rate for municipalities relative to municipal authorities

	(1)	(2)	(3)	(4)	(5)	(6)
Baseline specification	0.291*** (0.0391)	0.281*** (0.0387)	0.262*** (0.0382)	0.299*** (0.0426)	0.292*** (0.0419)	0.274*** (0.0408)
RC1: Cluster standard errors at the municipality level	0.291*** (0.0383)	0.281*** (0.0399)	0.262*** (0.0395)	0.299*** (0.0384)	0.292*** (0.0393)	0.274*** (0.0389)
RC2: Introduction of municipality FE	0.298*** (0.0481)	0.293*** (0.0470)	0.280*** (0.0453)	0.305*** (0.0492)	0.300*** (0.0481)	0.286*** (0.0460)
RC3: Omit all FE, either county or municipality	0.288*** (0.0387)	0.277*** (0.0385)	0.257*** (0.0382)	0.296*** (0.0433)	0.288*** (0.0427)	0.270*** (0.0416)
RC4: Include a proxy of the plan's asset allocation, the change in funding ratio between 2007 & 2009	0.292*** (0.0391)	0.281*** (0.0389)	0.261*** (0.0382)	0.300*** (0.0426)	0.292*** (0.0419)	0.274*** (0.0407)
RC5: Splitting the sample of municipal plans based on their (recalibrated) funding ratio in 2003						
DiD estimate for municipal plans with funding less than the median in 2003	0.392*** (0.0374)	0.420*** (0.0367)	0.417*** (0.0367)	0.398*** (0.0409)	0.412*** (0.0399)	0.414*** (0.0388)
DiD estimate for municipal plans with funding more than the median in 2003	0.191*** (0.0469)	0.149*** (0.0480)	0.115** (0.0474)	0.202*** (0.0504)	0.178*** (0.0514)	0.142*** (0.0507)

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Unless otherwise specified, all results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)—(6) replicate the patterns in cols. (1)—(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income. All regressions (except in RC1) cluster standard errors at the county level, with RC1 clustering at the municipal level.

Table A.5: Change in level of assets between any two consecutive years in the data computed at various percentiles in the distribution

Current year	Prior year	N	p1	p5	p10	p25	p50	p75	p90	p95	p99
2005	2003	1666	-18.54	-4.12	0.99	8.52	16.16	29.76	51.29	80.63	224.48
2007	2005	2015	-13.16	-1.35	3.31	8.50	15.34	24.91	39.79	54.13	99.82
2009	2007	2015	-30.86	-21.95	-15.58	-6.55	5.29	15.56	25.73	34.81	61.39
2011	2009	2015	-17.67	-6.08	-0.91	6.13	12.89	22.74	36.55	48.55	71.25
2013	2011	2015	-22.26	-8.11	-2.15	4.74	10.88	17.61	26.20	31.72	45.26

This table supports the decision to use the change in level of assets as a proxy for the asset mix (particularly, the equity allocation) of the plan. We can also see that the distributions are different – in particular, there is a much longer left tail in the distribution of change in level of assets between 2007 and 2009 than what we see in other years. Recognize that also because of use of asset smoothing techniques, not all of the drop in the level of assets between 2007 and 2009 may be reflected in the level of assets reported in 2009.

Fig. A.1: Change in actuarial assets for each two-year period between 2003 and 2013 (e.g. 2003 and 2005, 2005 and 2007, etc.)

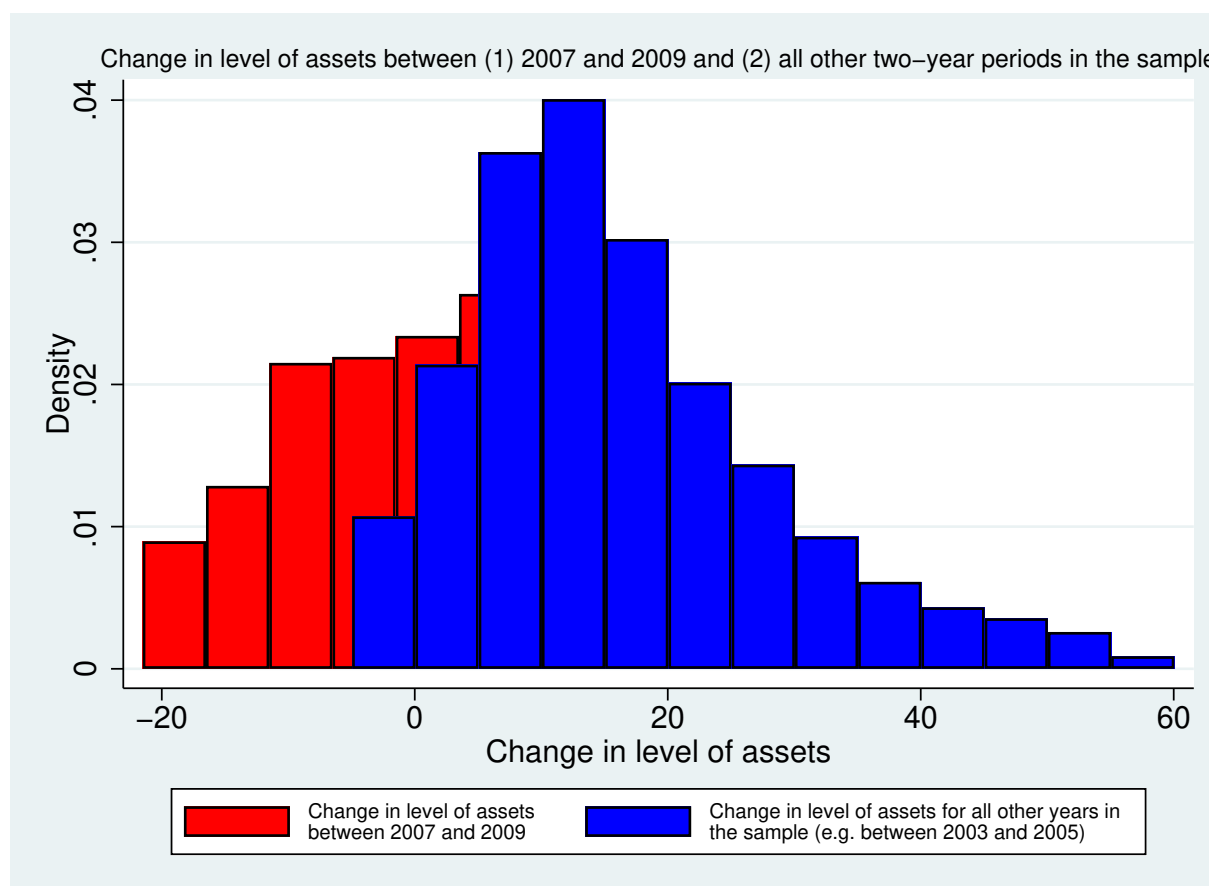


Table A.6: Change in funding ratio between any two consecutive years in the data computed at various percentiles in the distribution

Current year	Prior year	N	p1	p5	p10	p25	p50	p75	p90	p95	p99
2005	2003	1672	-69.20	-28.79	-17.65	-8.40	-0.60	8.19	19.99	30.37	69.25
2007	2005	2012	-43.40	-19.09	-11.18	-3.33	1.33	7.93	15.55	22.44	45.61
2009	2007	2012	-77.69	-41.20	-31.05	-17.72	-7.51	0.41	7.16	12.08	31.07
2011	2009	2013	-41.51	-17.85	-11.86	-5.45	0.00	6.04	14.70	21.04	45.34
2013	2011	2011	-37.99	-17.00	-11.94	-5.93	-1.13	3.20	8.86	14.47	39.29

This table supports the decision to use the change in funding ratio as a proxy for the asset mix (particularly, the equity allocation) of the plan. We can also see that the distributions are different – in particular, there is a much longer left tail in the distribution of change in funding ratios between 2007 and 2009 than what we see in other years.

Fig. A.2: Change in funding ratio for each two-year period between 2003 and 2013 (e.g. 2003 and 2005, 2005 and 2007, etc.)

